

Using GPU causes Data to not be found

<https://github.com/Qiskit/qiskit-aer/issues/662>

ROW 9

Using GPU causes Data to not be found #662

New issue

Closed

#717



shreychowdhary opened on Mar 15, 2020 · edited by shreychowdhary

Edits ...

Informations

- Qiskit Aer version: 0.4.1
- Python version: 3.6
- Operating system: ubuntu

What is the current behavior?

It throws the following exceptions after running the code for a while:

```
Traceback (most recent call last):
  File "/home/dino/.local/lib/python3.6/site-packages/qiskit/result/result.py", line 265, in _get_experiment_data
    exp = next(result for result in self.results
StopIteration

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
  File "nvqe.py", line 211, in <module>
    fidelity,energy,iterations = vqe.run()
  File "nvqe.py", line 162, in run
    ret = optimizer.optimize(num_vars=self.num_params, objective_function=self.objective_func, initial_point=self.initial_point, **self.options)
  File "/home/dino/.local/lib/python3.6/site-packages/qiskit/aqua/components/optimizers/spsa.py", line 139, in optimize
    **self.options)
  File "/home/dino/.local/lib/python3.6/site-packages/qiskit/aqua/components/optimizers/spsa.py", line 190, in _optimize
    cost_plus = obj_fun(theta_plus)
  File "nvqe.py", line 99, in objective_func
    total = self.objective_fun_rec(params,self.N,None)
  File "nvqe.py", line 117, in objective_fun_rec
    total += self.objective_fun_rec(params,N-1,str(i),p)
  File "nvqe.py", line 119, in objective_fun_rec
    total += self.objective_fun_rec(params,N-1,pauli_indexes+str(i),p)
  File "nvqe.py", line 110, in objective_fun_rec
    expected, neg = self.calculate_expected(result.get_counts(qc),pauli_indexes)
  File "/home/dino/.local/lib/python3.6/site-packages/qiskit/result/result.py", line 176, in get_counts
    exp = self._get_experiment(experiment)
  File "/home/dino/.local/lib/python3.6/site-packages/qiskit/result/result.py", line 270, in _get_experiment
    key)
qiskit.exceptions.QiskitError: 'Data for experiment "circuit15860" could not be found. '
```

Steps to reproduce the problem

<https://gist.github.com/shreychowdhary/beb0bc8c9ad2d2908536a2019b72073f>
while running using qiskit-aer-gpu

What is the expected behavior?

Its should run through without any errors which it does if I do not use the statevector_gpu backend option.

Suggested solutions

None



shreychowdhary added **bug** on Mar 15, 2020

shreychowdhary changed the title ~~Using GPU cause Data to not be found~~ Using GPU causes Data to not be found on Mar 15, 2020

atilag assigned [doichanj](#) on Mar 16, 2020

Assignees

[doichanj](#)

[hhorii](#)

Labels

bug

Type

No type

Projects

No projects

Milestone

Aer 0.5.1

Closed on Apr 28, 2020, 100% complete

Relationships

None yet

Development

[fix memory leak in gpu simulator](#)

Qiskit/qiskit-aer

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Participants

[shreychowdhary](#) [doichanj](#) [hhorii](#) [atilag](#) +1

shreychowdhary added **bug** on Mar 15, 2020

shreychowdhary changed the title ~~Using GPU cause Data to not be found~~ Using GPU causes Data to not be found on Mar 15, 2020

atilag assigned doichanj on Mar 16, 2020

chriselectic assigned hhorii on Apr 14, 2020

chriselectic added this to the Aer 0.5.1 milestone on Apr 15, 2020

jwoehr on Apr 16, 2020 · edited by jwoehr

Edits Contributor ...

My code is using master branch (updated daily, it's 2020-04-15 today) I'm experiencing the same error.

- The BasicAer simulators return results.
- The Aer statevector gpu simulator and density matrix gpu simulator both return the error shown by @shreychowdhary due to no results.



chriselectic on Apr 20, 2020

Member ...

@jwoehr, are any GPU tests passing for you? (If you have the git repo checked out run `python -m unittest test.terra.backends.test_qasm_simulator_statevector.TestQasmSimulatorStatevectorThrustGPU`)

@hhorii @doichanj Have you been able to look into this yet?



jwoehr on Apr 20, 2020 · edited by jwoehr

Edits Contributor ...

@chriselectic I have master branch checked out, yes, but can't get that command to work for me ... what directory should I be in to execute it?

```
$ python -m unittest test.terra.backends.test_qasm_simulator_statevector.TestQasmSimulatorStatevectorThrustGPU
E
=====
ERROR: terra (unittest.loader._FailedTest)
=====
ImportError: Failed to import test module: terra
Traceback (most recent call last):
  File "/usr/lib/python3.6/unittest/loader.py", line 153, in loadTestsFromName
    module = __import__(module_name)
ModuleNotFoundError: No module named 'test.terra'
```



jwoehr on Apr 22, 2020

Contributor ...

@hhorii I got a notification from you about a similar error on the Mac but I can't find your message here ...



chriselectic on Apr 22, 2020

Member ...

@jwoehr That command should be executed in the `qiskit-aer` git repo directory (you may need to have installed in executable mode `pip install -e .` for it to work properly)





Contributor ...



Collaborator ...

GPU simulator may have memory leak.

The status of execute error was ERROR: std::bad_alloc: cudaErrorMemoryAllocation: out of memory.



 hhorii mentioned this on Apr 26, 2020

[fix memory leak in gpu simulator #717](#)

Collaborator ...

@jwoehr thank you for your report. I was able to fix it. I believe this fix will be included in the next release (5.2).



Contributor ...

Thank you @hhorii I pull `master` branch daily, I'll watch for your pull request to be accepted!

  **atilag** closed this as **completed** in **#717** on Apr 27, 2020

Edits Contributor

@atilag @hhorii : I pulled the latest and I don't see any difference in behavior:

```
File ~/home/jax/work/QISKit/qiskit_dev_venv4/lib/python3.6/site-packages/qiskit_terra-0.14.0-py3.6-linux-x86_64-qiskit-experiments/_experiment_data.py:79:
    return self._get_experiment(experiment).data.to_dict()
File ~/home/jax/work/QISKit/qiskit_dev_venv4/lib/python3.6/site-packages/qiskit_terra-0.14.0-py3.6-linux-x86_64-qiskit-experiments/_experiment_data.py:79:
    key)
qiskit.exceptions.QiskitError: 'Data for experiment "circuit0" could not be found.'
```

Here's the job dictionary after the run:

```
{ 'args': ({'method': 'density_matrix_gpu'}, None, False),
  'backend': <QasmSimulator('qasm_simulator') from AerProvider()>,
  'fn': <bound method AerBackend.run_job of <QasmSimulator('qasm_simulator') from AerProvider()>>,
  'future': <Future at 0x3760a90ee58 state:finished returned:Result>
```

```
qiskit.exceptions.QiskitError: 'Data for experiment "circuit0" could not be found.'
```

Here's the job dictionary after the run:

```
{ '_args': ({'method': 'density_matrix_gpu'}, None, False),  
  '_backend': <QasmSimulator('qasm_simulator') from AerProvider(>,&br/>  '_fn': <bound method AerBackend.run_job of <QasmSimulator('qasm_simulator') from AerProvider(>>,&br/>  '_future': <Future at 0x7f69c9ccaa58 state=finished returned Result>,&br/>  '_job_id': '117d78d5-e76c-47c1-8251-fd38452173c2',  
  '_qobj': QasmQobj(qobj_id='9c1b6252-93bf-493c-9838-89035ca2dbb5', config=QasmQobjConfig(backend_options={
```



hhorii on Apr 28, 2020 · edited by hhorii

Edits Collaborator ...

@jwoehr Your first error was on `circuit15860`. This means that your 0-15859-th simulation were finished successfully and 15860-th simulation failed.

Your last error was `circuit0`. I believe that your simulator failed without any. Its reason may be that you did not build GPU simulator.

Could you tell me you specified `-DAER_THRUST_BACKEND=CUDA` or not to build the master?

Ex: `python setup.py bdist_wheel -- -DAER_THRUST_BACKEND=CUDA -- -j8`

I think you can find a hint of an error reason by checking `result` object)

```
result = execute(qc, backend, backend_options={"method": "statevector_gpu", "max_parallel_experiments": 1  
if (not result.success):  
    print(result)
```



jwoehr on Apr 28, 2020

Contributor ...

@hhorii

Ex: `python setup.py bdist_wheel -- -DAER_THRUST_BACKEND=CUDA -- -j8`

Ha! Did not realize I had to do that :)

Where is the build process for Aer GPU documented so I can point the next person who asks me to this step?



atilag on May 21, 2020

Member ...

Ouch, I just find out about this message.

And you are right, the install process is documented but not the build one.

I'll document it!

